Algebra Monsoon 2026 Lecture 20 Nilpotent groups

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Nilpotent groups

Def An extension of groups

$$1 \to N \to G \to Q \to 1$$

is called **central** if $N \subset Z(G)$.

Def. A group is **nilpotent** if it is

- commutative, or
- a central extension of a nilpotent group.
  (by a commutative group)

So every nilpotent group is solvable.
(by induction)

Example G_3 is not nilpotent

- it is not commutative
- $Z(G_3) = \{e\}$, so it is not a central extension of some other group.

Although, G_3 is solvable:

$$1 \to A_3 \to G_3 \to \mathbb{Z}^\times \to 1$$

is exact, with A_3 and $\mathbb{Z}^\times$ commutative.

More formally, a group G is said to be nilpotent if, in the sequence of groups

$$G_0 = G, \quad G_1 = G/Z(G), \quad G_{n+1} = G_n / Z(G_n)$$

there is an $n \geq 0$ such that

$$G_n = \{e\}.$$

(Compare with the def. of a solvable group: there is an $n \geq 0$ such that $G_n = \{0\}$ in the sequence

$$G_0 = G, \quad G_1 = D(G_0), \quad \dots, \quad G_{n+1} = D(G_n)$$

$\dots$)

Let p be a prime number.

Proposition Every p-group is nilpotent.

Proof By induction on n, where $o(G) = n$.

If $n = 1$, then G is commutative hence nilpotent.

Suppose $n \geq 1$. We have seen that $o(Z(G)) \geq p$, so $G/Z(G)$ is nilpotent by the inductive hyp. Therefore G is nilpotent, being a central extension of $G/Z(G)$ by $Z(G)$. $\square$